

Average slope:

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope at one point: also known as instant slope: is the derivative of the function

$$\frac{dy}{dx}$$

DERIVATIVES

Practice problems from Prof. Gilbert Strang's Lecture - Big Picture Derivatives

1. For $y = 2x^3$, what is the average slope $= \frac{\Delta y}{\Delta x}$ from $x = 1$ to $x = 2$?
2. What is the instant slope of $y = 2x^3$ at $x = 1$?
3. $y = x^n$ has $\frac{dy}{dx} = nx^{n-1}$. What is $\frac{dy}{dx}$ when $y(x) = \frac{1}{x} = x^{-1}$?
4. For $y = x^{-1}$, what is the average slope $\frac{\Delta y}{\Delta x}$ from $x = \frac{1}{2}$ to $x = 1$?
5. What is the instant slope of $y = x^{-1}$ at $x = \frac{1}{2}$?

Q1,2

$$y = 2x^3$$

given : two points: at $x = 1$ to $x = 2$

we first compute y values at the given points.

$$\begin{aligned} f(1) &= 2(1)^3 = 2 \\ f(2) &= 2(2)^3 = 16 \end{aligned}$$

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{16 - 2}{2 - 1} = \frac{14}{1}$$

then to find the instant slope , we differentiate the given equation.

$$\begin{aligned} f(x) &= 2x^3 \\ d/dx(x) &= 6x^2 \\ \text{at } x = 1, f'(x) &= 6(1)^2 = 6 \end{aligned}$$

q3.

$$\begin{aligned} f(x) &= \frac{1}{x} \\ f(x) &= 1.x^{-1} \\ f(x)' &= -1x^{(-1-1)} = -1x^{-2} \end{aligned}$$

Q4 & Q5

for

$$y = x^{-1}$$

what is the average slope from $x = 1/2$ to $x = 1$?

y values at the two given points are:

at $x = 1/2$

$$y = \left(\frac{1}{2}\right)^{-1} = \left(\frac{-1}{2}\right)^{-2} = 0.25$$

at $x = 1$

$$y = 1^{-1} = \frac{1}{1} = 1$$

Average slope =

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0.25}{1 - 0.5} = \frac{0.75}{0.5} = 1.5$$

Instant slope:

$$\begin{aligned} \frac{d}{dx} x^{-1} &= -1x^{-2} = \frac{-1}{x^2} \\ x &= \frac{1}{2} \\ \frac{-1}{\left(\frac{1}{2}\right)^2} &= \frac{-1}{0.25} = -4 \end{aligned}$$

Q6 Suppose the graph of $y(x)$ climbs up to its maximum at $x = 1$. Then it goes downward for $x > 1$.

6A. What is the sign of $\frac{dy}{dx}$ for $x < 1$ and then for $x > 1$?

6B. What is the instant slope at $x = 1$?

before $x=1$, that is when $x<1$, y increases, the slope is positive. at $x=1$, the slope is 0. and when $x>1$, slope is negative.

7 If $y = x$, write an expression for $\frac{\Delta y}{\Delta x}$ at any point x .

We see later that this $\frac{\Delta y}{\Delta x}$ approaches x

we need two points to calculate slope, but if the point is extremely close to the initial point such that the difference between the two points is almost 0, then such slope is called a derivative

in case of $\sin(x)$ let x be the first point and $x+h$ be the second point such that $h \rightarrow 0$

in that case $f(x) = \sin(x)$ and $f(x+h) = \sin(x+h)$

Change in y : $\sin(x+h)-\sin(x)$

change in x: $x+h - x = h$

slope: $\frac{\sin(x+h)-\sin(x)}{h}$

$$y(x) = x^2 - 2x \quad \frac{dy}{dx} = 0 \quad \text{at } x = -1, 0, 1. \quad \text{Find } y(-1), y(0), y(1).$$

3 Find the maximum height of $y(x) = -x^2 + 6x - 2$. **Solve** $\frac{dy}{dx} = 0$.

the equation can be re written as:

$$y(x) = -x^2 + 6x - 2$$

now we know its a down open parabola , the maximum point is

$$\frac{-b}{2a} = \frac{-6}{2(-1)} = \frac{-6}{-2} = 3$$

$$y(x)' = -2x + 6$$

$$-2x + 6 = 0, x = \frac{0 - 6}{-2} = 3$$

4 Find $\frac{d^2y}{dx^2}$ **to show that this parabola bends down.**

the second derivative is :

$$y(x)' = -2x + 6$$

$$y(x) = -x^2 + 6x - 2$$

$$\frac{d^2y}{dx^2} = -2$$

=,

5 For $y(x) = x^4 - 2x^2$ **show that** $\frac{dy}{dx} = 0$ **at** $x = -1, 0, 1$. **Find** $y(-1), y(0), y(1)$.

when $\frac{dy}{dx} = 0$, the slope is 0 , and those are the peaks (max and min) of the function's graph to find those points (max and min), we can do so by differentiating the function, factorizing it by setting it to 0.

$$\begin{aligned}
 y(x) &= x^4 - 2x^2 \\
 y(x)' &= 4x^3 - 4x \\
 4x^3 - 4x &= 0 \\
 4x(x^2 - 1) &= 0 \\
 4x = 0 \text{ or } x^2 - 1 = 0 &\quad (1) \\
 (x^2 - 1) &= (x - 1)(x + 1) \\
 (x - 1) = 0, x &= 1 \\
 (x + 1) = 0, x &= -1 \\
 \text{Therefore, } dy/dx = 0 &\text{ at } x = 1, -1
 \end{aligned}$$

6 Now $\frac{dy}{dx} = 4x^3 - 4x$. What is the second derivative $\frac{d^2y}{dx^2}$?

$$\frac{d^2y}{dx^2} = 12x^2 - 4$$

7 At a minimum point explain why $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$.

At minimum point the point is neither decreasing, nor increasing. So, it is horizontal without any upward or downward movement. Therefore dy/dx (the slope) is 0.

Likewise, the second derivative tells us how the slope is changing, if it is positive, the slope is changing in an upward/positive direction otherwise in a downward direction.

Exponential Functions

1 What is the derivative of $\frac{x^{10}}{10}$? $= \frac{x^{10}}{1} \cdot \frac{1}{10}$

derivative:

$$10x \cdot \frac{1}{10} x^{10-1} = \frac{10x}{10} = x$$

3 Why is $\frac{1}{x}$ the same as x^{-1} ?

exponential property recalled:

$$\begin{aligned}
 x^a \cdot x^b &= x^{a+b} \\
 x^a \cdot x^0 &= x^a \\
 x^{-a} \cdot x^a &= x^{-a+a} = x^0 = 1 \\
 x^{-a} &= \frac{1}{x^a} \\
 x^{-1} &= \frac{1}{x}
 \end{aligned}$$

4 Why is $x^{-1} = \frac{1}{x}$ between $\frac{1}{2}$ and $\frac{1}{3}$? Then 2 3.

We know, $x = 1 - x - \frac{1}{2}x^2 - \frac{1}{6}x^3$

when we substitute $x = 1$

$$1 = 1 - 1 - \frac{1}{2}1^2 - \frac{1}{6}1^3$$

$$1 = 1 - 1 - \frac{1}{2} - \frac{1}{6} = 2.12... () - \text{is between 2 and 3}$$

Since $1 = 1 - 1 - \frac{1}{2} - \frac{1}{6} = 2.1$,
we have $2 < 3$.

Then, $-x = \frac{1}{x} = \frac{1}{1 - x - \frac{1}{2}x^2 - \frac{1}{6}x^3}$ Taking reciprocals and flipping the inequalities,

$$\frac{1}{3} < \frac{1}{2} < \frac{1}{1}$$

$$-1 = \frac{1}{1} = \frac{1}{1 - 1 - \frac{1}{2}(1)^2 - \frac{1}{6}(1)^3} = \frac{1}{2.1...} < \frac{1}{3} \text{ and } \frac{1}{2}$$

5 Can you solve $\frac{dy}{dx} = y$ starting from $y = 3$ at $x = 0$?

we know, that

$$\begin{aligned} \frac{dy}{dx} &= y \\ y &= e^x \\ x = 0, y &= 1 \\ y &= 3 \quad y = 3^0 \end{aligned}$$

given

$$\frac{dy}{dx} = y = \frac{dy}{y} = dx$$

Why is $y = 3^x$ the right answer?

Because, the natural exponent is the only function which is the derivative of itself. hence $y = e^x$

6 Can you solve $\frac{dy}{dx} = 5y$ starting from $y = 1$ at $x = 0$? Why is $y = 5^x$ the right answer?

here $y = e^{5x}$
for function

$$5^x, \frac{d}{dx} 5^x = 5^x \ln 5 = 5^x \ln 5$$

7 Why does $\frac{\Delta y}{\Delta x}$ approach $\frac{dy}{dx}$ as Δx gets smaller?

using the difference quotient / Slope formula

$$\Delta y = \frac{dy}{dx} \Delta x$$

at $x = 0$ and h approaching 0, we can re write the above as:

$$\frac{dy}{dx} \bigg|_0 = \frac{0 - 0}{h} = \frac{-1}{h}$$

now as h approaches 0 ,

$$\frac{dy}{dx} \bigg|_0 = -1$$

Integrals

1 What functions $y(t)$ have the constant derivative $\frac{dy}{dt} = 7$?

$$\frac{dy}{dt} = 7 \Rightarrow y(t) = 7t + C$$

2 What is the area from 0 to t under the graph of $\frac{dy}{dt} = 7$?

$$\int_0^t 7 dt = 7t$$

3 From $t = 0$ to 2, find the integral $\int_0^2 7 dt =$

antiderivative = $7t$

from $t = 0$ to 2, $= 7(2) - 7(0) = 14 - 0 = 14$

4 What function $y(t)$ has the derivative $\frac{dy}{dt} = 6t$?

$$\frac{dy}{dt} = 6t \Rightarrow y(t) = 3t^2 + C$$

5 From $t = 0$ to 2, find area = integral $\int_0^2 (6t) dt$.

$$\int_0^2 (6t) dt =$$

$$3t^2 \bigg|_0^2 = (3(2)^2) - (3(0)^2) = 12 - 0 = 12$$

6 At this instant $t = 2$, what is $\frac{d}{dt} (3t^2)$?

the area function is $7t+3t^2$, the derivative is $s(t)$ itself which is $7+6t$
 at $t=2$, its $7+6(2)=7+12=19$

7 & 8

From 0 to t , the area under the curve $y = t$ IS NOT $y = t$. If t is small, the area must be small. But $t = 0$ has $y = 0 = 1$. From 0 to t , the correct area under $y = t$ is $y = t - 1$. The slope $\frac{dy}{dt}$ is _____ and now $y(0) =$

$$\frac{dy}{dt} = t$$

$$y(0) = 1 - 1 = 0$$

9 Notice y_0 in $(y_1 - y_0) (y_2 - y_1) (y_3 - y_2) = -y_0 y_3$ The sum of $\Delta y = \frac{\Delta y}{\Delta t} \Delta t$ becomes the integral of $\frac{dy}{dt} dt$ The area under (t) from 0 to t becomes $y(t) - y(0)$.

****18.01SC - Single Variable Calculus - Prof. David Jerison**

****Slope and Derivative****

a) Use the difference quotient definition of derivative to calculate the rate of change of the area of a disk with respect to its radius. (Your answer should be the circumference of the disk.)

area of a disk with radius r is :

$$f(r) = \pi r^2$$

now when r changes by " h "

$$= \frac{f(r+h) - f(r)}{h}$$

putting it in the area function:

$$= \frac{\pi (r+h)^2 - \pi r^2}{h} = \frac{\pi r^2 + 2\pi rh + \pi h^2 - \pi r^2}{h} = \frac{2\pi rh + \pi h^2}{h} = \pi (2r + h)$$

1C-3 Calculate the derivative of each of these functions directly from the definition.

a) $f(x) = 1/(2x - 1) = (2x - 1)^{-1}$
 $f'(x) = -1(2x - 1)^{-2} = -2(2x - 1)^{-2}$

d) $f(x) = 1/x = x^{-1}$
 $f'(x) = -1 x^{-2} = -\frac{1}{x^2}$

1C-4 Write an equation for the tangent line for the following functions:

a) $f(x) = 1/(2x - 1)$ at $x = 1$

$$f'(x) = -1/(2x - 1)^2 = -2/(2x - 1) = 1 \quad f'(1) = -2/(2(1) - 1) = -6$$

b) $f(x) = 2x^2 - 5x + 4$ at $x = a$

$$f'(x) = 4x - 5, \text{ at } x = a, f'(a) = 4(a) - 5$$

LIMITS & CONTINUITY

1D-1 Calculate the following limits if they exist. If they do not exist, then indicate whether they are ∞ , $-\infty$ or undefined.

a) $\lim_{x \rightarrow 0} \frac{4}{x - 1}$

substituting 0, we get :

$$\lim_{x \rightarrow 0} \frac{4}{x - 1} = \frac{4}{0 - 1} = \frac{-4}{1} = -4$$

c) $\lim_{x \rightarrow -2} \frac{4x^2}{x^2 - 2}$

$$\frac{4(-2)^2}{-2^2 - 2} = 16/0$$

denominator is 0

from the right, using -1.9 $\frac{4(-1.9)^2}{-1.9^2 - 2} = \frac{14.44}{0.1} = 144.4$

from the left side using -2.1 $\frac{4(-2.1)^2}{-2.1^2 - 2} = -176.4$

hence the left and right sides head somewhere different and not to a same point
the limit is undefined

d) $\lim_{x \rightarrow 2} \frac{4x^2}{2 - x}$

$$= \frac{4(2)^2}{2 - 2} = 16/0$$

the limit is not defined at 2.

from the right side, let's try plugging in 2.1.

$$= \frac{4(2.1)^2}{2 - 2.1} = \frac{4.41}{-0.1} = -44.1$$

plugging in 2.01

$$= \frac{4(2.01)^2}{2 - 2.01} = \frac{16.1604}{-0.01} = -1616.04.$$

so as x approaches 2 from the right, the value increases and goes towards $-\infty$

$$f) \lim_{x \rightarrow 2} \frac{4x^2}{x-2}$$

as x increases the numerator increases in a faster rate than the denominator

$$\begin{aligned} g) \lim_{x \rightarrow 2} \frac{4x^2}{x-2} - \frac{(4x)(x-2)}{x-2} \\ = \frac{4x^2}{x-2} - \frac{(4x)(x-2)}{x-2} = \frac{4x^2 - 4x^2 + 8x}{x-2} \\ = \frac{8x}{x-2} \end{aligned}$$

dividing by the highest degree term of denominator

$$= \frac{\frac{8x}{x}}{\frac{x-2}{x}} = \frac{8}{1 - \frac{2}{x}}$$

as x increases,, $2/x$ approaches 0

$$\text{so } \lim_{x \rightarrow 2} \frac{8}{1 - \frac{2}{x}} = \frac{8}{1 - 0} = 8$$

$$\begin{aligned} j) \lim_{x \rightarrow 2} \frac{x-2}{x^2-4} \\ = \frac{(x-2)(x+2)}{(x-2)(x+2)} \\ = \frac{1}{x+2} \\ = \frac{1}{4} = 0.25 \end{aligned}$$

1D-3 Identify and give the type of the points of discontinuity of each of the following:

$$a) \frac{x-2}{x^2-4}$$

finding the discontinuity:

setting the denominator to 0,

$$x^2 - 4 = 0$$

$$x = \pm 2$$

factoring the above equation:

$$\begin{aligned} \frac{(x-2)}{(x-2)(x+2)} \\ = \frac{1}{x+2} \end{aligned}$$

$(x-2)$ cancels so $x = 2$ is a fixable discontinuity
however, $x = -2$ is not fixable, its infinite discontinuity

b) $\frac{1}{x}$

at $x = 0$, , and every, $\sin$ is 0.

So at n , $\sin$ is 0

it doesn't cancel, so infinite discontinuity

c) $\frac{x^4}{x^3} = x$

it cancels completely and the discontinuity is completely removable

d) $f(x) = \begin{cases} x/a, & x \neq 0 \\ a-x, & x = 0 \end{cases}$

here, at $x = 1/a + a$ and $x = -1/a + 1$

both sides agree.

however, at $x = 0$ we don't have a defined function

at $x = 0$, $f(x) = a$ is a function that removes this discontinuity

e) $f'(x)$, for the $f(x) = \begin{cases} x/a, & x > 0 \\ a-x, & x < 0 \end{cases}$

$f'(x) = 1$ for $x > 0$ and -1 for $x < 0$

these are not agreeing. so there is a jump discontinuity

f) $(f(x))^2$, where $f(x) = \frac{d}{dx}x \rightarrow$

$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

$\frac{d}{dx}x = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$

$(f(x))^2 = \begin{cases} 1, & x > 0 \\ 1, & x < 0 \end{cases}$

the function $f(x) = 1$ at $x = 0$ removes discontinuity

Piecewise differentiability

1D-5 Define $f(x) = \begin{cases} ax, & x \leq 1 \\ x^2, & x > 1 \end{cases}$

a) Find all values of a , such that $f(x)$ is continuous.

b) Find all values of a , such that $f'(x)$ is continuous. (Be careful!)

1D-6 For each of the following functions, find all values of the constants a and b for which the function is differentiable.

$$a) f(x) = \begin{cases} x^2 - 4x + 1, & x \leq 0 \\ ax + b, & x > 0 \end{cases}$$

for a function to be differentiable, the left and right limits should match and the function must be continuous.

$$f(x) = \begin{cases} x^2 - 4x + 1, & x \leq 0 \\ ax + b, & x > 0 \end{cases}$$

the boundary point is at 0

so at $\lim_{x \rightarrow 0^-}$,

$$1 = b$$

the slopes of both the functions should match at $x = 0$

$$f'(x) = \begin{cases} 2x - 4, & x \leq 0 \\ a, & x > 0 \end{cases}$$

at $x = 0$, $a = 4$

hence, the function is continuous and differentiable at $a = 4$ and $b = 1$

1E-3 Find the points (x, y) of the graph $y = x^3 - x^2 - x - 2$ at which the slope of the tangent line is horizontal.

the slope is horizontal when it's 0 (i.e.) when there is no elevation.

$$y' = 3x^2 - 2x - 1$$

$$0 = 3x^2 - 2x - 1$$

$$3x^2 - 2x - 1$$

$$3x(x - 1) - 1(x - 1) = 0$$

$$(3x - 1)(x - 1) = 0$$

$$x = \frac{-1}{3} - 1$$

1H-3 Solve the following for y :

$$a) (y - 1)^x (y - 1) = 2x - x$$

$$(y - 1)^x (y - 1) = ((y - 1)^x (y - 1))$$

$$(y^2 - 1) = 2x - x$$

$$(y^2 - 1) - x = 2x$$

$$= \frac{(y-1)^x}{x} = 2x$$

$$= \frac{(y-1)^x}{x} = 2x$$

$$e^{\ln x} = x$$

$$\text{so } \frac{(y-1)^x}{x} = 2x$$

$$(y-1)^x = x^{2x}$$

$$y = x^{2x} + 1$$

$$b) (y-1) = x^2 (y-1)$$

$$\log(y+1) - \log(y-1) = x^2$$

$$\left(\frac{y+1}{y-1}\right) = x^2$$

removing log on both sides:

$$10^{(x^2)} = \frac{y+1}{y-1}$$

by substituting $k = 10^{x^2}$

$$k(y-1) = y+1$$

$$y = k(y-1) - 1$$

$$y = ky - k - 1$$

Linear and quadratic approximation

2A-1 Find the linearization of $a \cdot x$ at 0, by using (2), and also by using the basic approximation formulas. (Here a and x are constants; assume $a \neq 0$. Do not confuse this a with the one in (2), which has the value 0.)

$$\begin{aligned} f(x) &= (a \cdot x)^{1/2} \\ f'(x) &= \frac{1}{2}(a \cdot x)^{-1/2} \\ &= f(0) \quad f'(0) = (a)^{1/2} \quad \frac{1}{2}(a)^{-1/2} \end{aligned}$$

2A-6 Find a quadratic approximation to $\sqrt{x}$, for $x \geq 0$.

2A-12 Give the indicated type of approximation at the point indicated. (This is to be done after studying the log and exponential functions.)

$$a) \frac{x}{1-x} = \frac{1}{(1-x)^{-1}} \cdot x = (1-x)^{-1} \cdot x \text{ (quadratic, } x \geq 0)$$

quadratic approximation of terms:

$$\text{for } x = 1 \quad x = \frac{x^2}{2}$$

$$\text{for } (1-x)^{-1} \rightarrow (1-0)^{-1} - 1(1-x)^{-2} \cdot x = \frac{2(1-x)^{-3}}{2} \cdot x^2$$

using multiplication method:

the quadratic approximation of the equation :

$$(1 - x \cdot \frac{x^2}{2})((1-0)^{-1} - 1(1-0)^{-2} \cdot x = \frac{2(1-0)^{-3}}{2} \cdot x^2)$$

$$= (1 - x \cdot \frac{x^2}{2})(1 - x \cdot x^2)$$

$$= 1 - x^3 - \frac{x^2}{2} - \frac{x^3}{2} - \frac{x^4}{2}$$

$$= \frac{2x^3 + x^2 + x^4}{2}$$

dropping terms with degree higher than 2 :

we get:

$$= \frac{2x^2}{2}$$

c) $-x^2$ (quadratic, $x \geq 0$)

quadratic approximation = $f(0) + f'(0)x + \frac{f''(0)}{2}x^2$

$$f(0) = 1$$

$$f'(0) = -0^2 = -2(0)$$

$$f''(x) = (-x^2)' = -2x$$

$$f''(x) = -2 + (0)$$

$$\text{quadratic approximation} = 1 - x^2$$

a) $(2x)$, near 0

$$f(x) = \sin(2x)$$

$$f(0) = \sin(0) = 0$$

$$f'(x) = 2\cos(2x)$$

$$f'(0) = 2\cos(0) = 2$$

$$f''(x) = -2\sin(2x)$$

$$f''(x) = -2\sin(0) = 0$$

$$\text{quadratic approximation eqn: } 0 + 2x + 0 = 2x$$

OPTIMIZATION PROBLEMS

2C-1 Cut four identical squares out of the corners of a 12 by 12 inch piece of cardboard and fold the sides so as to make a box without a top. Find the size of the corner square that maximizes the volume of the box.

volume of the box after cutting out squares out of the corner. of size x

*volume of a cube/ cuboid: length.width *height*

$$\text{height} = x$$

$$\text{length and width are the same} = 12 - 2x$$

$$\text{so the volume of the cube is of the equation: } (x)(12-2x)(12-2x) = x(12-2x)^2$$

$$x(12^2 - 2 \cdot 12 \cdot 2x + 2(12)(2x) - 4x^2)$$

$$= x(144 - 4x^2 - 4x) = 144x - 4x^3 - 4x^2 = 4x^3 - 4x^2 - 144x$$

the critical points are:

$$4x^3 - 4x^2 - 144x = 0$$

$$4x(x^2 - 12x - 36) = 0$$

$$4x(x^2 - 6x - 6x - 36) = 0$$

$$4x(x(x - 6) - 6(x - 6)) = 0$$

$$4x(x - 6)^2 = 0$$

$$x = 6 \text{ or } x = 0$$

COMMENT: THIS QUESTION IS TO BE RE DONE

2C-2 You are asked to design a rectangular barnyard enclosing 20,000 square feet with fencing on three sides and the wall of a long barn on the fourth. Find the shortest length of fence needed.

the area of a rectangular fence would be $l \cdot b$.

the area = $lb = 20000$

$l = 20000/b$

the cost of total fencing would be $l+2b = c$ (assuming the wall is the bigger side)

substituting l : $20000^{-1} 2 = c$

derivative: $-20000^{-2} 2 =$

setting it to 0:

$$\frac{-20000}{2} 2 = 0$$

$$-20000 2^2 = 0$$

$$2^2 = 20000$$

$$2 = 10000$$

$$= / - 10000 = 100$$

b cannot be negative, so $b = 100$

at $b = 100$, $l = 20000/100$

$l = 200$

the total fencing required = $200 + 100 + 100 = 400$

2C-3 What is the largest value of the product xy , if x and y are related by $x + 2y = a$, where a is a fixed positive constant?

given $x + 2y = a$

$x = a - 2y$

$$xy = (a - 2y)y = ay - 2y^2$$

derivative:

$$a - 4y = 0$$

$$-4y = -a$$

$$4y = a$$

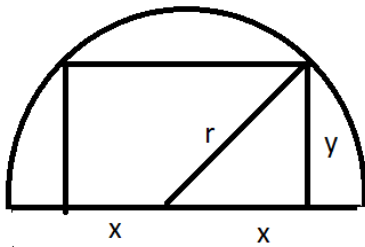
$$y = a/4$$

$$xy = a \frac{a}{4} - 2 \left(\frac{a}{4} \right)^2$$

$$xy = \frac{a^2}{4} - 2 \left(\frac{a^2}{16} \right)$$

$$xy = \frac{a^2}{4} - \left(\frac{a^2}{8} \right) = \frac{2a^2 - a^2}{8} = \frac{a^2}{8}$$

2C-7** Find the dimensions of the rectangle of largest area that can be inscribed in a semicircle of radius .



for rectangle inscribed in a semi circle, we can formulate that :

$$x^2 + y^2 = r^2$$

$$x^2 + y^2 = 2^2$$

$$x^2 = 2^2 - y^2$$

$$x = \sqrt{2^2 - y^2}$$

area of rectangle = $2xy$

substituting x value $\rightarrow$

$$2^2 - y^2 \cdot y$$

$$2y \cdot (2^2 - y^2)^{\frac{1}{2}}$$

differentiating this equation to find the maxima and minima

$$\frac{d}{dy} [2y(2^2 - y^2)^{\frac{1}{2}}] = 0 \Rightarrow 2(2^2 - y^2)^{\frac{1}{2}} - y^2(2^2 - y^2)^{-\frac{1}{2}} = 0$$

$$= -y^2(2^2 - y^2) = -(2^2 - y^2)$$

$$= -y^2 = -\frac{(2^2 - y^2)}{2^2 - y^2}$$

$$y^2 = \frac{1}{2^2 - y^2}$$

$$y = \sqrt{\frac{1}{2^2 - y^2}}$$

{this answer needs to be re worked}